

Qatar-1b

R-2752

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110813 · created 2026-09-02T23:36:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455786.79609 +/- 0.00041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.01 +/- 0.055
Transit depth (Rp/Rs)^2	0.0095 +/- 0.11 [%]
Orbital Inclination (inc)	88.72 +/- 2.95
Airmass coefficient 1 (a1)	38.61 +/- 0.32
Airmass coefficient 2 (a2)	-3.0 +/- 1.18
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	#2 - [410, 53]
Optimal Aperture	67.98
Optimal Annulus	119.98
Transit Duration (day)	0.00409 +/- 0.00048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.01 ± 0.055 vs 0.1463 (deviation 2.48σ)
Duration fit vs published	0.00409 d vs 0.0692 d (deviation 135.58σ)
Mid-time O-C	1.0 min
Depth SNR	0.2
Residual scatter	0.82 %

Light curve

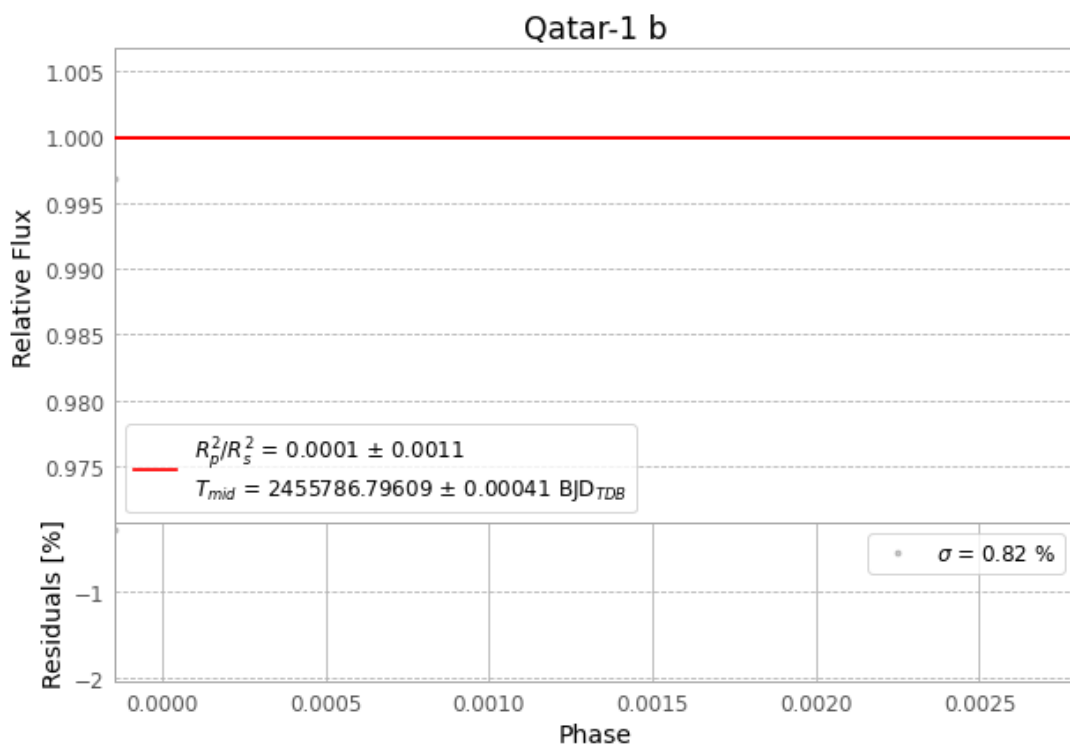


Figure 1 — FinalLightCurve_Qatar-1 b_2011-08-12.png (SHA-256 0653c9b01d715714...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a3d3ede984098553...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1110813052412.FITS → Qatar-1110813085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110813.log (e554b383d55e...), FinalLightCurve_Qatar-1 b_2011-08-12.png (0653c9b01d71...), FinalLightCurve_Qatar-1 b_2011-08-12.csv (0125a1fc447b...)

Compute resources

Wall time 834.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:36Z.